

NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering
Project Documentation

Course Title: Mobile Computing LAB

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Project Name: Medical Report Analyzer

Document: AI Integration Report

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Medical Report Analyzer

MedInsight is an AI-powered web application designed to help users understand complex medical reports in simple and accessible language. The system analyzes uploaded medical reports, identifies important findings and abnormal values, generates a plain-language summary, and provides an interactive AI assistant for follow-up questions.

The application uses Google Gemini as its primary AI model. The AI functionality addresses the meaningful problem of medical report comprehension rather than being included only for demonstration purposes. The system converts medical information into understandable explanations while encouraging users to consult qualified medical professionals for diagnosis and treatment.

1. Team Information

Student ID	Name	Role	Task Distribution
11220320959	Hassan A.K Azad Tushin	Team Leader	AI Integration
11220320899	Farjana Easmin Minka	AI Integration Lead & Database Manager	Documentation, Database Management, Testing
11220320982	Juena Tabassum	Frontend Developer	Frontend Development: React.js
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2. Selected AI Platform

Google Gemini

The project uses Google Gemini as the selected AI platform for intelligent medical report analysis and conversational assistance. Gemini was selected because it provides the natural-language understanding, structured response generation, context-aware question answering, and plain-language explanation capabilities required by the project.

Component	Technology
AI Platform	Google Gemini
AI Model	Gemini 3.5 Flash

Backend	Node.js + Express.js
Frontend	React + Vite
Database	PostgreSQL / Supabase
API Architecture	REST API
Authentication	JWT

3. AI Model Used

Google Gemini 3.5 Flash

Google Gemini 3.5 Flash is used as the main AI model in MedInsight. The model receives processed medical report content and generates structured information that can be consumed by the application.

The main AI-generated outputs are:

- Plain-language summary
- Key findings
- Abnormal result identification
- Risk level assessment
- Recommendations
- Context-aware chatbot responses

The report-analysis response is structured into fields such as `summary_text`, `key_findings`, `abnormal_flags`, `risk_level`, and `recommendations`. The AI is not intended to replace a doctor or provide a definitive diagnosis; its purpose is to help users understand the information contained in their reports.

4. AI Features Implemented

4.1 AI Medical Report Analysis

The main AI feature analyzes the content of a medical report using Gemini. The system receives the report, processes its content, sends the relevant content to Gemini, analyzes laboratory markers and values, identifies abnormal findings, generates a structured response, stores the analysis, and displays the result.

4.2 Plain-Language Medical Summary

Gemini converts complex medical terminology into simpler language that can be understood by non-medical users. This directly addresses the project's primary objective of improving medical report comprehension.

4.3 Key Findings Extraction

Gemini identifies important laboratory markers and their corresponding values and returns the findings in a structured format so that the frontend can display them consistently.

4.4 Abnormal Result Identification

Gemini identifies values that are outside the expected range and creates abnormal flags. Each abnormal finding contains the marker name, reported value, status, and an explanation/reason.

4.5 Risk Level Assessment

The AI provides an overall informational risk classification using Low, Medium, or High. This classification does not constitute a medical diagnosis.

4.6 AI Recommendations

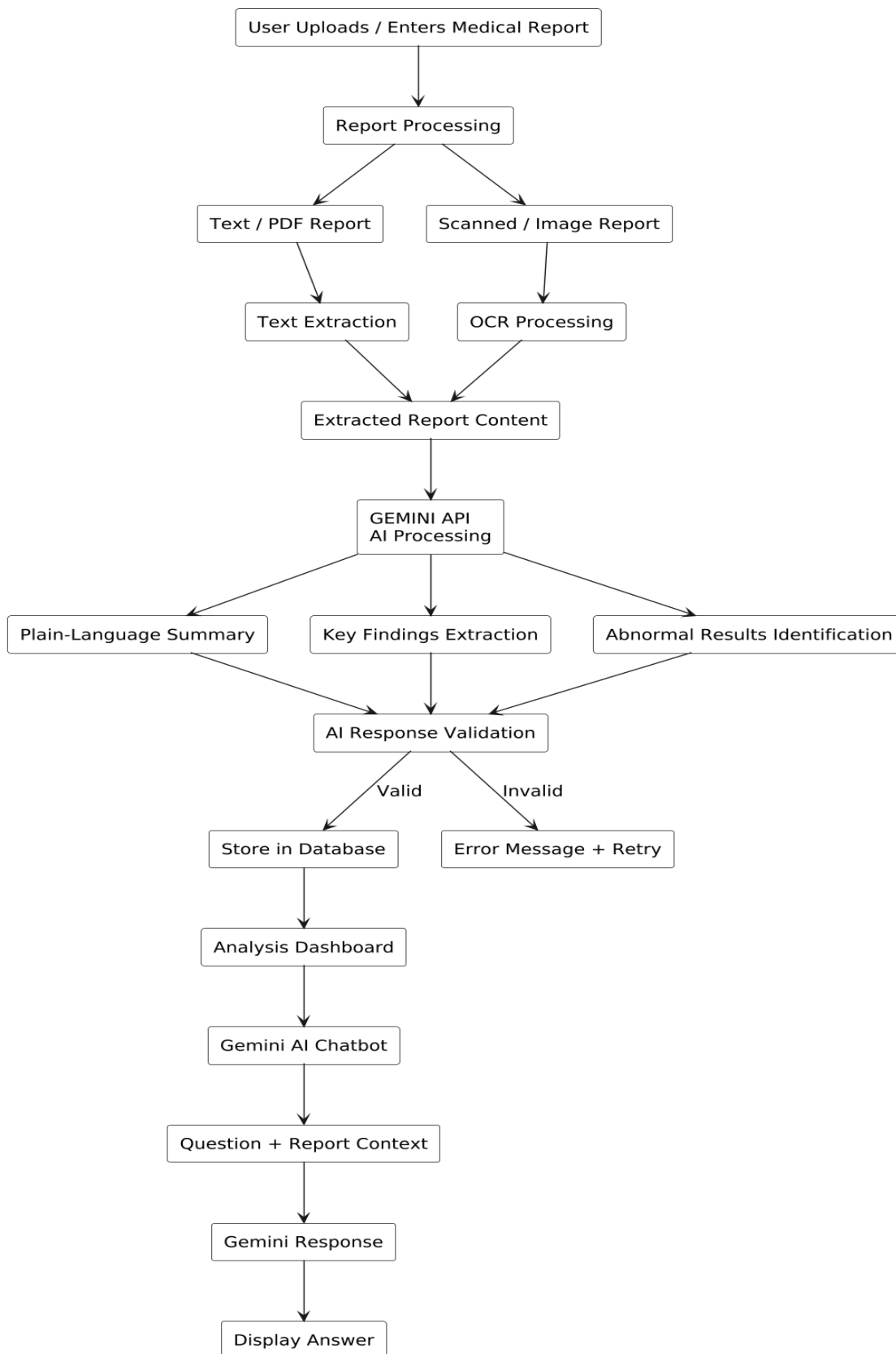
Gemini can provide general follow-up or lifestyle recommendations based on the analyzed report. These are informational and users are advised to consult a qualified doctor for diagnosis or treatment.

4.7 AI Medical Chat Assistant

The AI Assistant allows users to ask follow-up questions about their medical report. The chatbot receives the user's question together with the selected report context and sends it to Gemini. Conversations are stored for future reference.

5. AI Workflow Design

The complete AI workflow of MedInsight follows the sequence below:



5.1 User Uploads / Enters Medical Report

The user uploads a medical report or enters the report information into the system.

5.2 Report Processing

The system processes the submitted report and determines how its content should be extracted.

5.3 Text / PDF Report

For digital or text-based reports, the existing text content is processed directly.

5.4 Scanned / Image Report

For scanned or image-based reports, OCR processing is used to extract readable text from the document.

5.5 Text Extraction

The system extracts machine-readable text from the submitted report.

5.6 OCR Processing

OCR converts text contained in scanned documents or images into machine-readable report content.

5.7 Extracted Report Content

The extracted content is prepared and passed to the AI system for further analysis.

5.8 Gemini API — AI Processing

The extracted report content is sent to the **Google Gemini API**, where Gemini analyzes the medical information according to the application's prompts.

5.9 Plain-Language Summary

Gemini generates a simplified summary of the medical report so that users can understand the information more easily.

5.10 Key Findings Extraction

Gemini identifies the important medical and laboratory findings contained in the report.

5.11 Abnormal Results Identification

Gemini identifies potentially abnormal values or findings that may require the user's attention.

5.12 AI Response Validation

The application validates the Gemini response to ensure that the generated output is complete, properly structured, and usable.

5.13 Valid Response — Store in Database

If the response is valid, the generated analysis is stored in the database for future access.

5.14 Invalid Response — Error Message + Retry

If the AI response is invalid or the AI service encounters an error, the system displays an appropriate error message and allows the process to be retried.

5.15 Analysis Dashboard

The validated AI-generated analysis is displayed on the user's medical report analysis dashboard.

5.16 Gemini AI Chatbot

The user can interact with the Gemini-powered chatbot to ask questions about their medical report.

5.17 Question + Report Context

The user's question is combined with the relevant report context so that Gemini can provide a context-aware answer.

5.18 Gemini Response

Gemini processes the question and report context and generates an appropriate response.

5.19 Display Answer

The generated response is displayed to the user through the AI chatbot interface.

6. Prompt Engineering Examples

6.1 Report Analysis — System Prompt

You are MedInsight AI, an expert medical report analyzer.

Your job is to analyze patient lab reports and provide clear, accurate, and easy-to-understand results.

RULES:

- Always respond with ONLY valid JSON — no markdown, no explanation outside JSON.
- Be accurate but use plain language a non-doctor can understand.
- Flag anything outside normal ranges.
- Never diagnose — only analyze and suggest consulting a doctor.

RESPONSE FORMAT (strict JSON):

```
{
  "summary_text": "2-3 sentence plain language summary",
  "key_findings": [
    { "marker": "name", "value": "value with unit", "status":
      "Normal|High|Low" }
  ],
  "abnormal_flags": [
    { "marker": "name", "reason": "plain language explanation" }
  ],
  "risk_level": "Low|Medium|High",
  "recommendations": ["recommendation 1", "recommendation 2"]
}
```

6.2 Report Analysis — User Prompt

Analyze this medical lab report and return structured JSON:

[REPORT TEXT INSERTED HERE]

Return ONLY the JSON object as specified. No extra text.

6.3 Chat Assistant — System Prompt

You are MedInsight AI, a friendly and knowledgeable medical assistant.

You help patients understand their lab report results in simple, clear language.

RULES:

- Answer only health and medical report related questions
- Use simple language — avoid complex medical jargon
- Always recommend consulting a real doctor for diagnosis or treatment
- Be empathetic and reassuring
- Keep answers concise (2-4 sentences)
- If asked something unrelated to health, politely redirect

You have access to this patient's report context: {REPORT_CONTEXT}

6.4 Prompt Engineering Strategy

Version	Approach	Result	Selected
V1	Simple text prompt	Inconsistent JSON	No
V2	JSON structure in prompt	Better structure	No
V3	Strict JSON with examples	More reliable	No
V4	System prompt + structured response configuration	Consistent JSON output	Yes
V5	Higher temperature	More creative but less consistent	No
V6	Lower temperature	More accurate and consistent	Yes

6.5 Expected AI Output Examples

Example 1 — Report Analysis Output:

```
{
  "summary_text":
    "Your blood test shows mild anemia and elevated
    cholesterol. Fasting glucose is in the pre-diabetic range. Kidney
    and liver function are normal. Please discuss with your doctor.",

  "key_findings":
    [
      { "marker": "Hemoglobin", "value": "11.2 g/dL", "status": "Low" },
      { "marker": "Cholesterol", "value": "228 mg/dL", "status": "High" }
    ],
  "abnormal_flags":
    [
      { "marker": "Hemoglobin",
        "reason": "Below normal. May cause fatigue. Iron-rich diet advised." }
    ],
  "risk_level": "Medium",

  "recommendations": ["Follow up with doctor in 4 weeks",
    "Reduce saturated fat intake"]
}
```

Example 2 — Chat Response Output:

User: "What does my cholesterol level mean?"

AI: "Your cholesterol of 228 mg/dL is above the recommended limit of 200 mg/dL. This increases your cardiovascular risk over time. Reducing saturated fats and increasing exercise can help bring it down. Please discuss medication options with your doctor if lifestyle changes aren't enough."

7. AI Response Handling Strategy

Since the application depends on an external Gemini API, the system handles situations in which the AI does not return a valid response. The AI service is designed so that an AI failure does not crash the report-upload process.

Scenario	Handling Strategy
Successful response	Validate, process, store, and display the AI result.
Empty response	Detect empty/very short output and use safe fallback handling.
Invalid JSON	Catch parsing error and use fallback handling.
Invalid API key	Display an appropriate configuration/API error.
Rate limit	Detect HTTP 429 and handle without crashing.
Timeout	Abort the request and provide a retry message.
Network failure	Catch the request failure and provide fallback handling.
Gemini unavailable	Continue application operation using fallback response.
Database save failure	Mark the report as failed and log the error.

8. Current Development Progress

#	Development Task	Status	Details
1	Gemini API Integration	Completed	Gemini integrated through backend AI service.
2	AI Report Analysis	Completed	Report content processed by Gemini.
3	Summary Generation	Completed	Plain-language summary generated.
4	Key Findings Extraction	Completed	Important report markers extracted.

5	Abnormal Result Detection	Completed	Abnormal values flagged.
6	Risk Level Assessment	Completed	Low/Medium/High classification implemented.
7	AI Recommendations	Completed	General follow-up recommendations generated.
8	AI Chat Assistant	Completed	Report-context-based Q&A implemented.
9	Chat History	Completed	Conversations stored in database.
10	AI Response Validation	Completed	AI output validation implemented.
11	Error Handling	Completed	Timeout, API, network, rate-limit and invalid-response handling.
12	OCR / Report Processing	Implemented	Report content processing for supported report formats.
13	Frontend AI Integration	Completed	AI results and assistant connected to React interface.
14	Database Integration	Completed	AI summaries and chat history stored in PostgreSQL/Supabase.

8.1 Current Development Status

AI Integration: Completed. The current AI implementation includes the Gemini-powered report analyzer, structured AI output, validation, database storage, report-specific chatbot, chat history, and error-handling mechanisms.

12. Current Limitations

12.1 Dependence on Gemini API

The system depends on the availability of the Gemini API. API errors, network problems, or rate limits may temporarily affect AI analysis.

12.2 AI Response Accuracy

Gemini responses may not always be completely accurate and should be treated as informational rather than a medical diagnosis.

12.3 Limited Medical Context

The AI mainly analyzes information available in the uploaded report and does not have complete access to the user's medical history or clinical examination.

12.4 Report Format Limitations

Poor-quality, unclear, or unusually formatted reports may affect text extraction and the quality of AI analysis.

12.5 API Usage Limitations

The use of an external Gemini API introduces usage limits and potential costs as the number of users and AI requests increases.

12.6 Limited Personalization

Current AI responses are mainly based on the uploaded report and user questions, with limited personalization based on long-term medical history.

13. Future Improvements

13.1 Advanced AI Analysis

Improve Gemini-based analysis to provide deeper interpretation of multiple medical parameters and their relationships.

13.2 Medical History Integration

Allow users to maintain their medical history so the AI can consider previous health information.

13.3 Report Comparison

Add automatic comparison between previous and current reports to identify changes in health indicators.

13.4 Admin Panel

Add an admin panel for managing users, reports, AI usage, and overall system activity.

13.5 Improved AI Chatbot

Enhance the Gemini chatbot with better conversational memory, contextual understanding, and follow-up question handling.

13.6 Multi-Language Support

Add support for multiple languages, particularly **Bangla and English**, to make medical information more accessible.

13.7 Enhanced Security and Scalability

Improve data security, access control, and system scalability to support a larger number of users and protect sensitive medical information.

6. GitHub Repository Link

Main Repository : <https://github.com/Tushin-vai/CSE-4204-8D-T02>

Frontend Branch: https://github.com/Tushin-vai/CSE-4204-8D-T02/tree/Frontend_Juena-Tabassum/frontend

Backend Branch: https://github.com/Tushin-vai/CSE-4204-8D-T02/tree/Backend_faikanewazsilmi/backend

AI Integration Branch: <https://github.com/Tushin-vai/CSE-4204-8D-T02/tree/AI-Integration-Farjana-Yasmin-Minka>